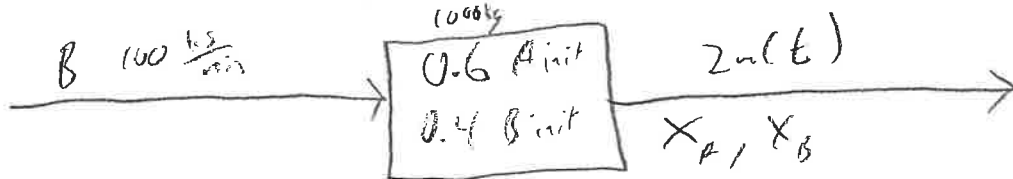
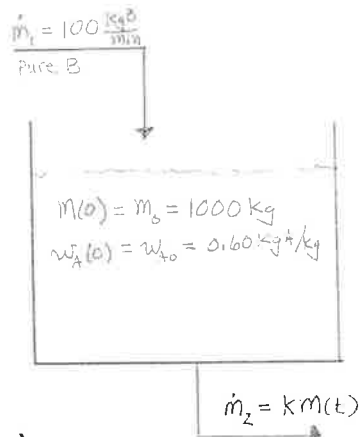


To Receive Full Credit, You Must Show All Your Work

Initially at $t = 0$, a mixing tank contains $m(0) = 1000$ kg of a binary liquid mixture of species A and B. The initial mass fraction of species A is $w_A(0) = 0.60$. The inlet flow rate is $\dot{m}_1 = 100$ kgB/min of pure B. The outlet flow rate is given by $\dot{m}_2 = k * m(t)$ where $k = 2.0 \text{ min}^{-1}$ and $m(t)$ is the mass in the tank as a function of time. Assume the tank is well mixed throughout the process.

- (a) At steady state, how much mass, in kg, is left in the tank? That is, when $t \rightarrow \infty$, what is $m(t \rightarrow \infty)$?
 (b) Derive a general expression for $m(t)$. How long does it take for $m(t)$ to drop to 100 kg?
 (c) Derive a general expression for $m_A(t)$, the mass of A in the tank. At the time calculated in part (b), what is w_A , the mass fraction of A in the tank?



a) $100 = 2m$
 50 kg at steady state

$$\frac{dm}{dt} = 100 - 2m$$

$$\frac{dm}{dt} = 2(50 - m)$$

$$\frac{1}{50 - m} dm = 2 dt$$

$$\frac{1}{m - 50} dm = -2 dt$$

$$\ln(m - 50) = -2t + C$$

$$m - 50 = C e^{-2t}$$

$$950 = C$$

$$m(0) = 1000$$

$$C =$$

$$m = 950 e^{-2t} + 50$$

$$100 = 950 e^{-2t} + 50$$

$$\frac{1}{e^{-2t}} = e^{2t}$$

$$t = 1.47 \text{ min}$$

$$(c) \frac{dm_A}{dt} = (-2m)w_A$$

$$m_A(t) = w_A^T m_{sys}$$

$$\frac{d(w_A m_{sys})}{dt} = -2 w_A m_{sys}$$

$$m \frac{dw_A}{dt} + w_A \frac{dm}{dt} = -2 w_A m$$

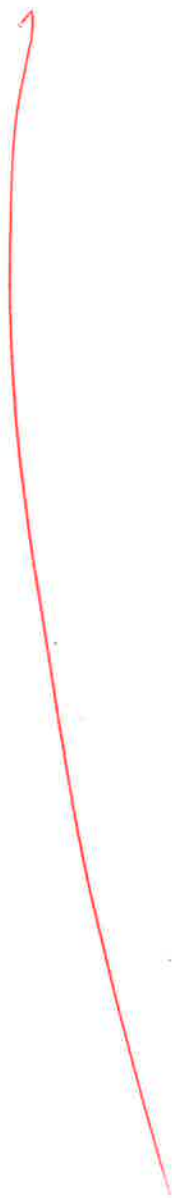
$$m \frac{dw_A}{dt} = -2 w_A m - w_A \frac{dm}{dt}$$

$$m \frac{dw_A}{dt} = w_A (-2m - \frac{dm}{dt})$$

$$m \frac{dw_A}{dt} = w_A (-1900 e^{-2t} - 100 - (0.6 - 2m))$$

$$\frac{1}{w_A} \frac{dw_A}{dt} = \frac{-1900 e^{-2t} - 200 + 2m}{m}$$

(next page)



NAME Tamir Stuart

Some useful integrals:

$$\int (a + bx) dx = \frac{1}{2b} (a + bx)^2$$

$$\int \frac{dx}{(a + bx)} = \frac{1}{b} \ln|a + bx|$$

$$\int \frac{dx}{x(a + bx)} = -\frac{1}{a} \ln \left| \frac{a + bx}{x} \right|$$

$$\int \frac{x dx}{(a + bx)} = \frac{1}{b^2} (a + bx - a \ln|a + bx|)$$

$$\int \frac{dx}{(a + bx)^2} = \frac{-1}{b(a + bx)}$$

$$\int \frac{(A + Bx)}{(a + bx)} dx = \frac{(bA - aB) \ln|a + bx| + bBx}{b^2}$$

$$\int \frac{dx}{(a + bx)(A + Bx)} = \frac{1}{(bA - aB)} \ln \left| \frac{a + bx}{A + Bx} \right|$$

$$\ln X_A = \int_0^t$$

~~ln(X_A)~~

$$\frac{1}{X_A} \frac{dX_A}{dt} = \frac{-1900e^{-2t}}{950e^{-2t} + 10} - \frac{200}{950e^{-2t} + 10} + 2$$

$$\ln(X_A) \Big|_{0.6}^{1.47} = \int_0^{1.47} \left(-\frac{1900e^{-2t}}{950e^{-2t} + 10} - \frac{200}{950e^{-2t} + 10} + 2 \right) dt$$

$$\ln V_f - \ln(0.6) = -0.6396$$

$$V_f = 0.32$$

How did you do this integral?
let $u = e^{-2t}$
 $du = -2e^{-2t} dt$
numerically?

